Experimenting with Classifiers in Weka

This article explains how to design and run classification algorithms on the well-known Weka platform—the open source machine learning software that can be accessed through a GUI, terminal, or a Java API. It is aimed at students, faculty members and researchers interested in machine learning.

Machine learning algorithms possess the capabilities of self-learning and improvisation without explicitly being programmed, and have been getting very popular in recent years. Supervised learning is one of the machine learning tasks by which algorithms usually learn with labelled data. Classification is one of the supervised learning tasks which learns from labelled data and predicts the categorical class labels. There are many tools and libraries available for experimenting with machine learning algorithms on various types of data sets. One of the most popular among these is Weka, which was developed at the University of Waikato, New Zealand.

Weka contains many tools. It is open source software built in Java and offers functions for data visualisation, preprocessing, regression, clustering, classification and association rules. Here we will discuss some well-known algorithms offered by Weka. Shown below is a step-by-step illustration of how classification is performed on an available data set.

1. Download Weka from *https://waikato.github.io/weka-wiki/downloading_weka/* and complete the installation process as per the given guidelines.
2. After successful installation, start the Weka GUI. It offers four options to work with, as described in Figure 1.
3. Select the tab named 'Explorer' on the GUI. It will open the Explorer

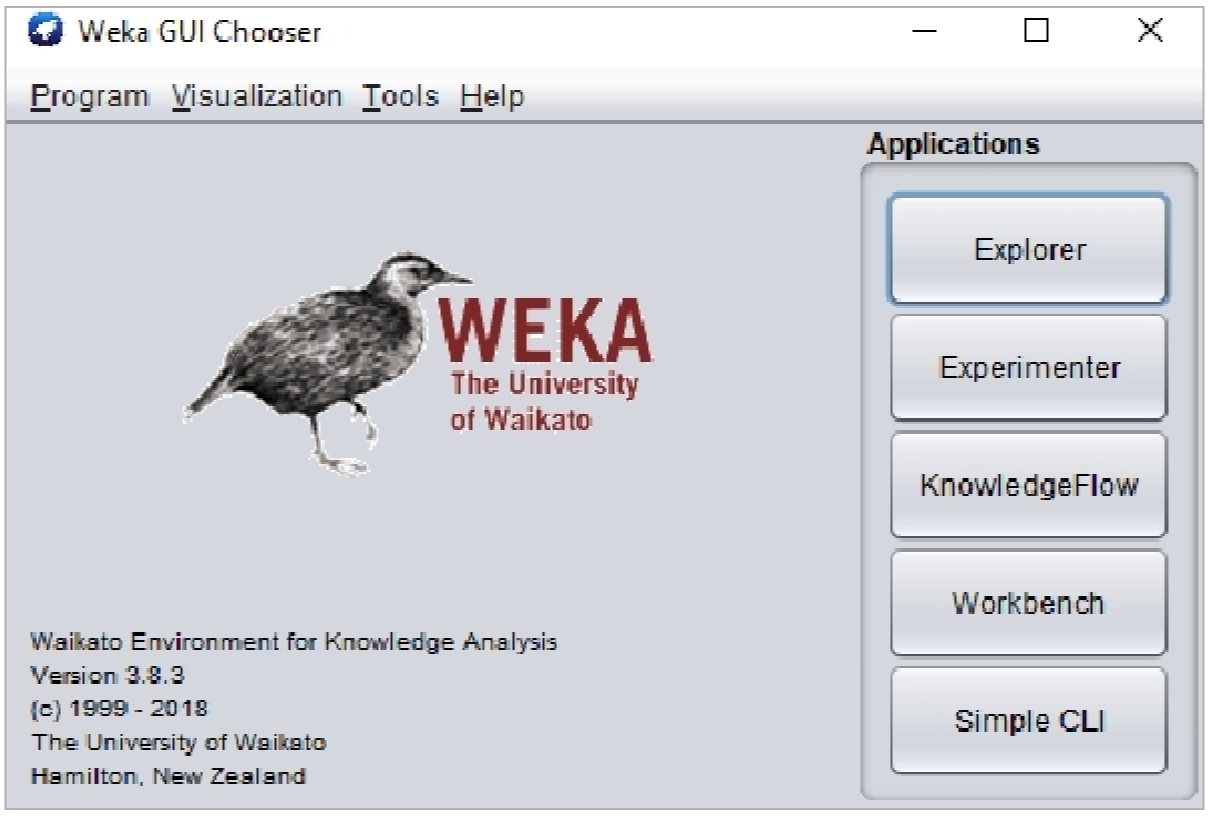

Figure 1: The Weka GUI selector

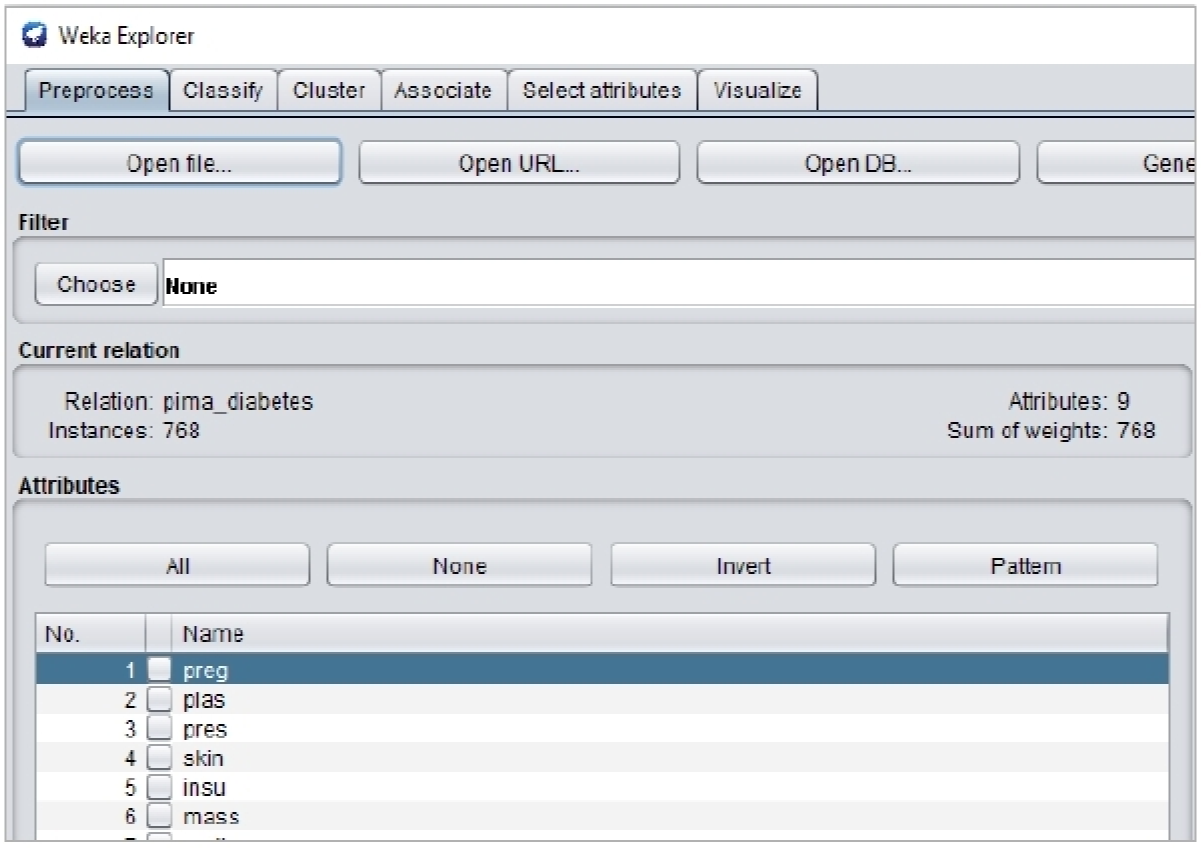

Figure 2: Preprocessing in Weka